Gregory Polyakov

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Education

PhD **University of Tübingen & Max Planck Research School**, Computer Science — Tübingen, Germany, Sept 2024 – present

- Health-NLP Lab, Supervisor: Prof. Dr. Carsten Eickhoff.
- Affiliated Scholar, International Max Planck Research School for Intelligent Systems (IMPRS-IS).
- Research topics: Interpretability of Deep Learning Models for NLP and IR tasks.

MSc **Moscow Institute of Physics and Technology**, Applied Mathematics and Computer Science — Moscow, Russia, Sept 2021 – July 2024

- GPA: 3.35 / 4.00
- Research topics: Sparse Retrieval, Multimodal Retrieval, Tool Calling in LLMs.
- Thesis: "Approaches to Training Language Models for External Function Calling: Using Self-Generated Data for Better Error Detection and Recovery".

BSc **Moscow Institute of Physics and Technology**, Applied Mathematics and Computer Science — Moscow, Russia, Sept 2017 – July 2021

- GPA: 3.48 / 4.00
- Thesis: "Usage of MCMC Methods in Deep Learning".

Experience

Huawei Noah's Ark Lab, NLP Research Engineer (full-time) — Moscow, Russia, Nov 2022 – Sept 2024, 1 year 11 months

- Investigated LLM tool usage capabilities. Developed a fine-tuning strategy for error recovery, leading to a publication at the ACL 2025 REALM Workshop.
- Conducted research on multimodal (Text-Video) retrieval, contributing to video retrieval post-processing techniques published at SIGIR 2023.
- Designed and co-implemented the end-to-end NLU pipeline for a production car voice assistant in Russian.

Huawei Noah's Ark Lab, NLP Research Intern (full-time) — Moscow, Russia, Oct 2021 – Nov 2022, 1 year 2 months

- Co-developed and optimized the reranking algorithm for a large-scale product search engine in Russian.
- Improved the reranker model quality by 5% and reduced its inference latency by 1.5x by adapting novel architectural (ColBERTer) and distillation approaches (TinyBERT).

Publications

Interpretability Analysis of Arithmetic In-Context Learning in Large Language Models — 2025
G. Polyakov, C. Hepting, C. Eickhoff, A. Bahrainian
(EMNLP 2025)

Towards Best Practices of Axiomatic Activation Patching in Information Retrieval — 2025
G. Polyakov, C. Chen, C. Eickhoff
(SIGIR 2025)

ToolReflection: Improving Large Language Models for Real-World API Calls with Self-Generated Data — 2025
G. Polyakov, I. Alimova, D. Abulkhanov, I. Sedykh, A. Bout, S. Nikolenko, I. Piontkovskaya
(ACL 2025 REALM Workshop)

Sinkhorn Transformations for Single-Query Postprocessing in Text-Video Retrieval 2023
K. Yakovlev, G. Polyakov, I. Alimova, A. V. Podolskiy, A. Bout, S. I. Nikolenko, I. Piontkovskaya
(SIGIR 2023)

Scholarships

Presidential Grant Scholarship Feb 2018
Selected as one of 1,200 recipients nationwide for this highly competitive prize recognizing high academic merit. Included a full 4-year stipend supporting undergraduate research and studies.

Teaching

Teaching Assistant: Modern Search Engines University of Tübingen
Summer Semesters 2025 and 2026. Conducted hands-on tutorials, evaluated assignments, and mentored student projects.

Teaching Assistant: Introduction to Python Programming University of Tübingen
Winter Semesters 2024/25 and 2025/26. Conducted hands-on tutorials and evaluated student assignments.

Languages

Russian: Native

English: C1 (IELTS Band Score 7)